



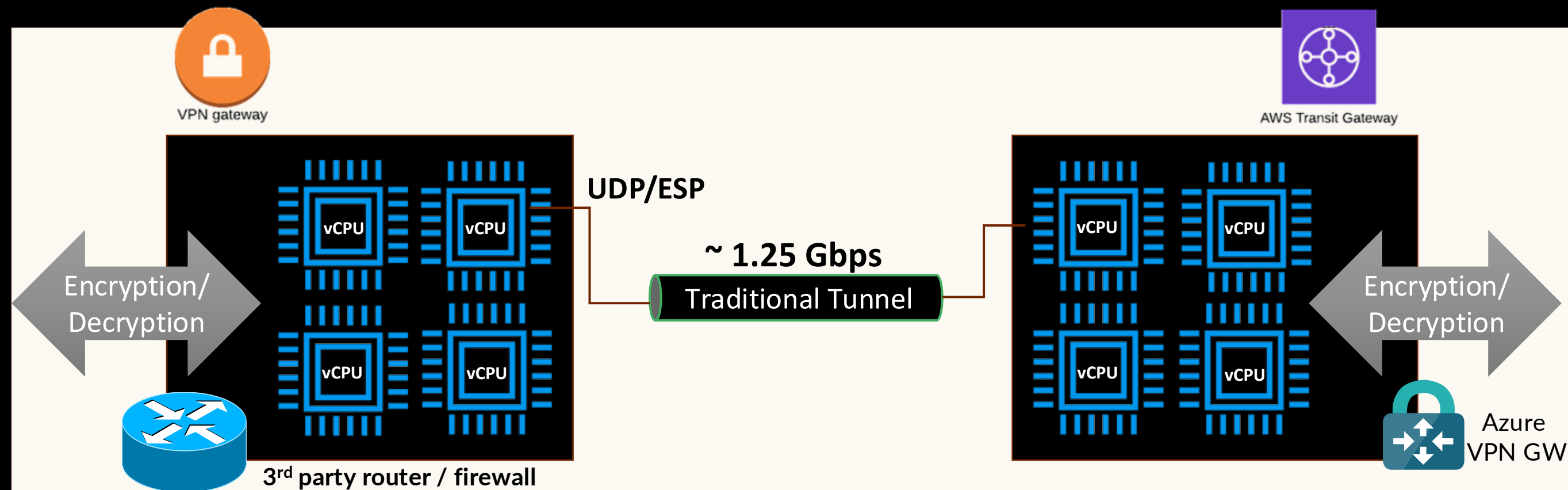
High-Performance Encryption (HPE)

aka Insane Mode



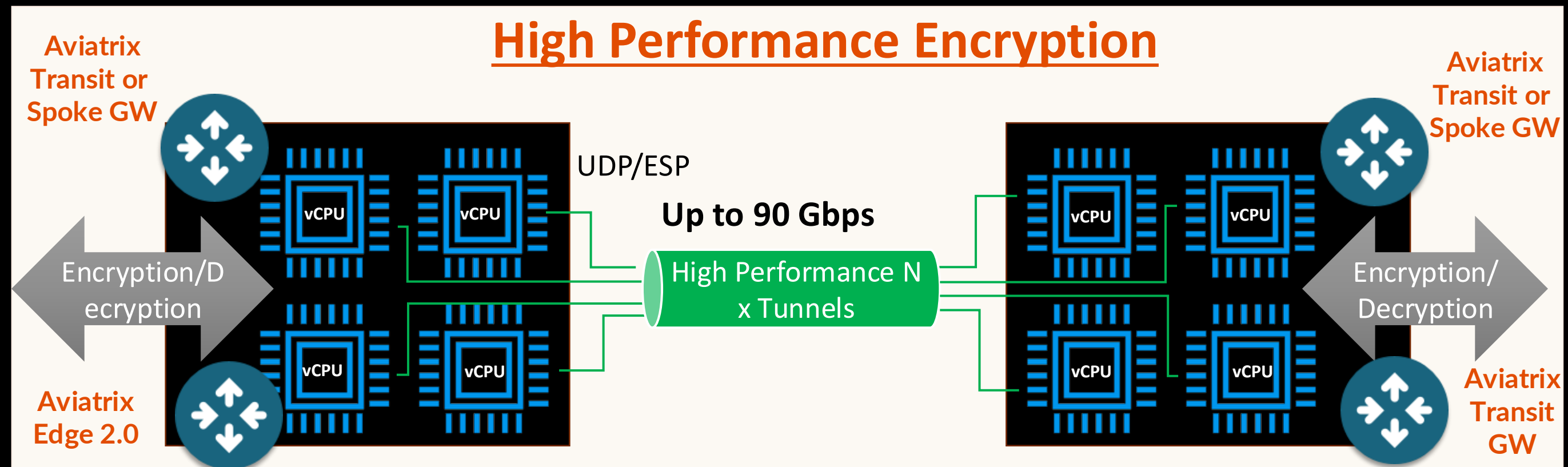
Without the Aviatrix CNSF: Encryption / IPSec Performance Limitations

- ❑ All software-based IPsec VPN solutions have maximum performance of 2Gbps depending on ciphers used
- ❑ Software Routers use single core and establish only one tunnel
- ❑ Packet can only use single core despite availability of multiple cores



Solution: Aviatrix High-Performance Encryption (HPE)

- ❑ Aviatrix Controller automatically builds multiple tunnels between Aviatrix devices
- ❑ Uses all available CPU cores
- ❑ IPsec encryption performance can be up to 90 Gbps



- High-performance encryption, previously branded as “Insane Mode.”

Solution: Aviatrix High-Performance Encryption (HPE)

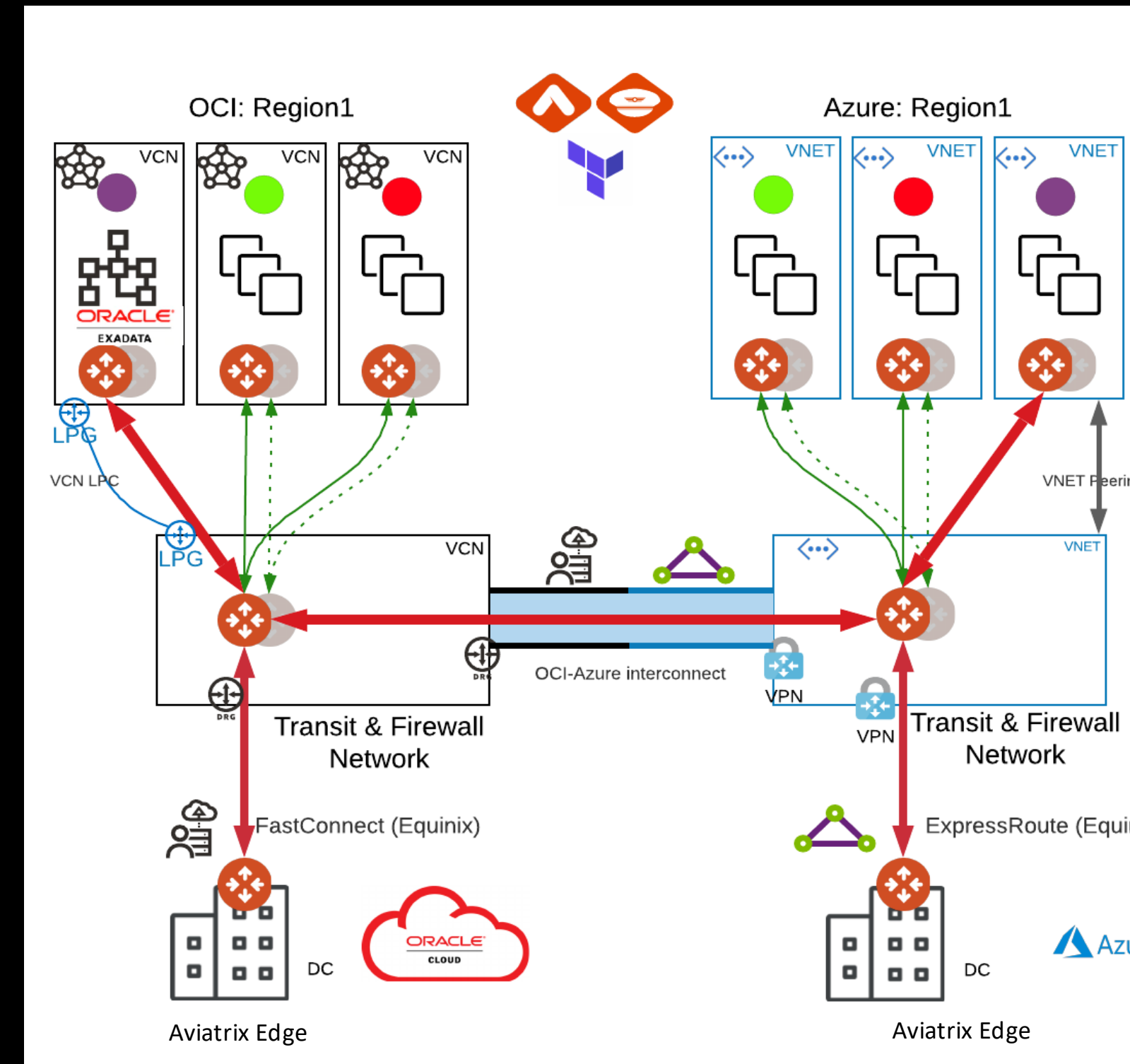
Cloud Provider	Instance SIZES that support HPE
AWS	t3 (spoke), t3a (spoke), c5 (spoke and transit), c5n (spoke and transit), c6in (spoke and transit)
Azure	Standard (except for B1ms, B2s, B4ms, B8ms, D1_v2, D2_v2, DS1_v2, DS2_v2, D2s_v3, D4s_v3, F2s_v2, F4s_v2)
GCP	n1-standard (except for standard-1 and standard-2), n1-highcpu (except for highcpu-2)
OCI	All instance sizes

❏ **Caveat:** The gateway instance size determines how many tunnels are created.

HPE – Use Cases

- Between the Cloud (over DirectConnect, ExpressRoute, FastConnect, Cloud Interconnect) to the DC via:
 - **Aviatrix Edge**
- Between networks in one cloud (same or different regions)
 - Automatic VPC/VNet/VCN peering to build required underlay
- Between networks in different clouds
 - Requires private underlay (e.g., Equinix, Epsilon, Megaport, OCI-Azure Interconnect)
 - Over Public Internet

- Aviatrix Edge will be discussed in Site2Cloud module



HPE peering – Public or Private IPs?

❑ HPE in the same cloud:

- Will use *CSP-native peering* so the tunnels will be built over private IPs.

❑ HPE across different clouds:

- Supported over private circuits (Direct Connect, Express Route, Cloud Interconnect, Fast Connect).
- Supported over internet (AWS, Azure, GCP, OCI).

Next: ActiveMesh, Failover & Control Plane

